Evaluation of Large Language Models for Discovery of Gene Set Function

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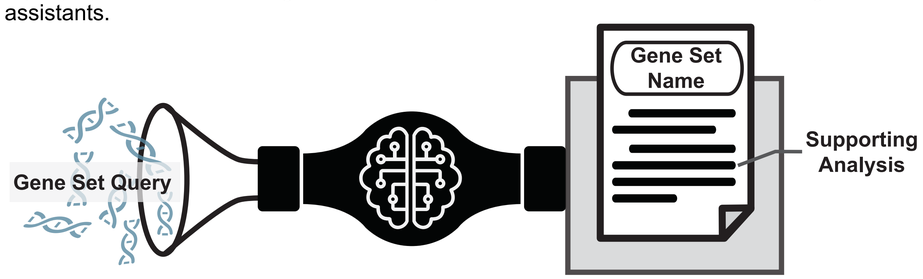
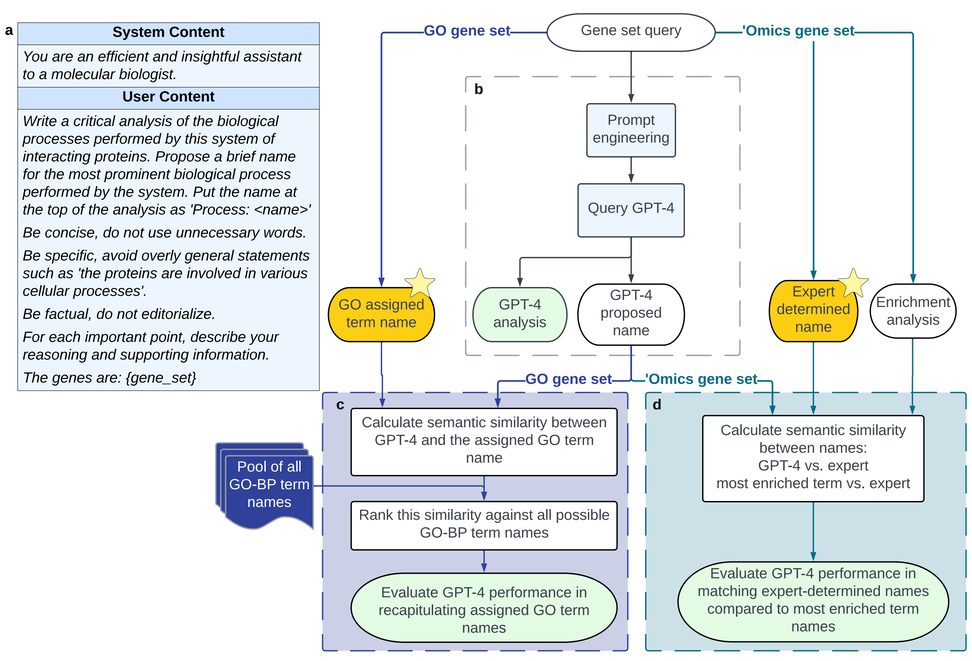
Abstract

Gene set analysis is a mainstay of functional genomics, but it relies on manually curated databases of gene functions that are incomplete and unaware of biological context. Here we evaluate the ability of OpenAI's GPT-4[1], a Large Language Model (LLM), to develop hypotheses about common gene functions from its embedded biomedical knowledge. We created a GPT-4 pipeline* to label gene sets with names that summarize their consensus functions, substantiated by analysis text and citations. Benchmarking against named gene sets in the Gene Ontology[2], GPT-4 generated very similar names in 50% of cases, while in most remaining cases it recovered the name of a more general concept. In gene sets discovered in 'omics data, GPT-4 names were more informative than gene set enrichment, with supporting statements and citations that largely verified in human review. The ability to rapidly synthesize common gene functions positions LLMs as valuable functional genomics assistants.

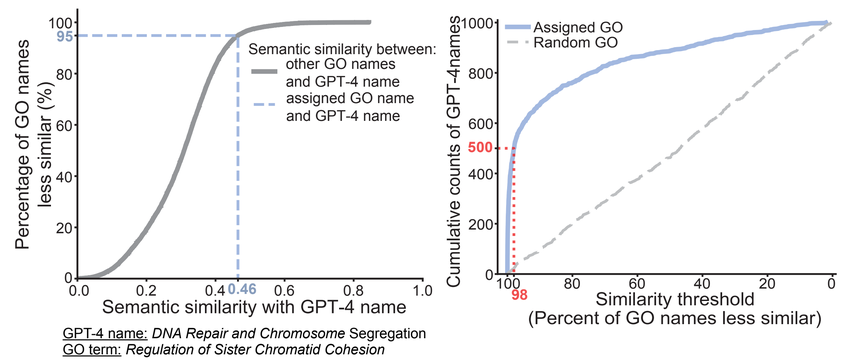


Use and evaluation of GPT-4 for functional analysis of gene sets



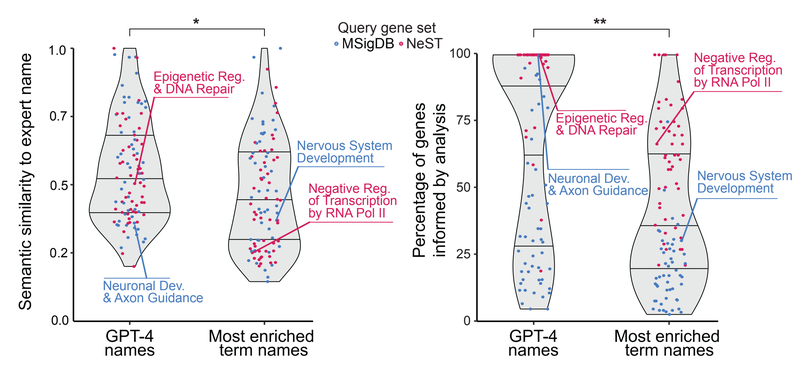
a, The prompt provided to GPT-4 for its input. The specific list of genes is inserted into the {gene_set}.
b, Generation of a proposed name for a gene set, with an accompanying analysis.
c, Benchmarking against Gene Ontology gene sets.
d, Benchmarking against 'omics gene sets (MSigDB hallmark gene sets[3], NeST cancer systems[4]).

GPT-4 is capable of generating descriptive gene set names



Best and worst GPT-4 names for GO terms by semantic similarity.

GO Name (# of genes)	GPT-4 Name	Semantic Similarity	Rank among all GO terms (n = 12,214)
GPT-4 Names with Highest Similarity to GO Names			
Mitochondrial tRNA Modification (9)	Mitochondrial tRNA Modification	1.00	1
Acute-Phase Response (48)	Acute Phase Response	0.99	1
Branched-Chain Amino Acid Catabolic Process (21)	Branched–Chain Amino Acid Catabolism	0.99	1
Homocysteine Metabolic Process (14)	Homocysteine Metabolism	0.99	1
Tryptophan Catabolic Process (9)	Tryptophan Catabolism	0.98	1
GPT-4 Names with Lowest Similarity to GO Names			
Ovulation From Ovarian Follicle (10)	Regulation of Cell Proliferation and Apoptosis	0.16	11,431
Positive Regulation of Prostaglandin Biosynthetic Process (7)	Inflammation Regulation	0.16	11,945
Positive Regulation of Protein Import Into Nucleus (39)	Hedgehog Signaling Pathway	0.15	11,324
Positive Regulation of Norepinephrine Secretion (4)	Oxytocin Signaling and Regulation	0.14	11,964
Positive Regulation of Histone H4 Acetylation (6)	DNA Repair and Transcription Regulation	0.13	11,887



Text rich analysis provides supporting statements

Representative analysis for protein interaction clusters (NeST:34 "Histone Modification").

Genes (n = 30)	BARD1 BCL11B BRD4 CHD4 CREBBP DNMT1 DNMT3A DNMT3B E2F1 EMSY EZH2 HDAC1 HDAC2 HDAC3 HIST3H3 ING1 IRF2 KDM5A MTA1 PALB2 PARP1 PHF6 PRDM1 RB1 RUNX1 SMARCA1 TERT TP53 ZBTB2 ZNF217		
Analysis Method	**Assigned Name**	**Semantic Similarity**	**# Genes Covered**
GPT-4	Epigenetic Regulation and DNA Repair	0.50	30
Gene set enrichment	Negative Regulation Of Transcription By RNA Polymerase I	0.25	20

GPT-4 Analysis Paragraph

The primary biological process performed by this system of interacting proteins is epigenetic regulation and DNA repair. The proteins involved in this system play crucial roles in modulating gene expression, chromatin remodeling, and maintaining genomic stability.

1. Chromatin remodeling and histone modification: CHD4, SMARCA1, and ING1 are involved in chromatin remodeling, which is essential for regulating gene expression. HDAC1, HDAC2, and HDAC3 are histone deacetylases that remove acetyl groups from histones, leading to chromatin condensation and transcriptional repression. CREBBP is a histone acetyltransferase that adds acetyl groups to histones, promoting transcriptional activation. EZH2 is a histone methyltransferase that catalyzes the addition of methyl groups to histone H3, leading to gene silencing. KDM5A is a histone demethylase that removes methyl groups from histone H3, allowing for gene activation.

2. DNA methylation: DNMT1, DNMT3A, and DNMT3B are DNA methyltransferases that add methyl groups to cytosine residues in DNA, leading to gene silencing. This process is crucial for maintaining genomic stability and regulating gene expression.

3. Transcriptional regulation: E2F1, IRF2, PRDM1, RB1, RUNX1, TP53, and ZBTB2 are transcription factors that bind to specific DNA sequences and regulate gene expression. BCL11B and ZNF217 are also involved in transcriptional regulation through their interactions with other transcription factors and chromatin modifiers.

4. DNA repair and genomic stability: BARD1, BRD4, EMSY, MTA1, PALB2, and PARP1 are involved in DNA repair pathways, such as homologous recombination and non-homologous end joining. These proteins help maintain genomic stability by repairing DNA damage and preventing mutations.

5. Telomere maintenance: TERT is the catalytic subunit of telomerase, which is responsible for maintaining telomere length and preventing cellular senescence.

6. Protein-protein interactions: PHF6 and HIST3H3 are involved in protein-protein interactions that modulate the activity of other proteins in this system.

In summary, this system of interacting proteins plays a critical role in epigenetic regulation and DNA repair, ensuring proper gene expression and maintaining genomic stability.

Acknowledgements

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*Code available at https://github.com/idekerlab/llm_evaluation_for_gene_set_interpretation

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